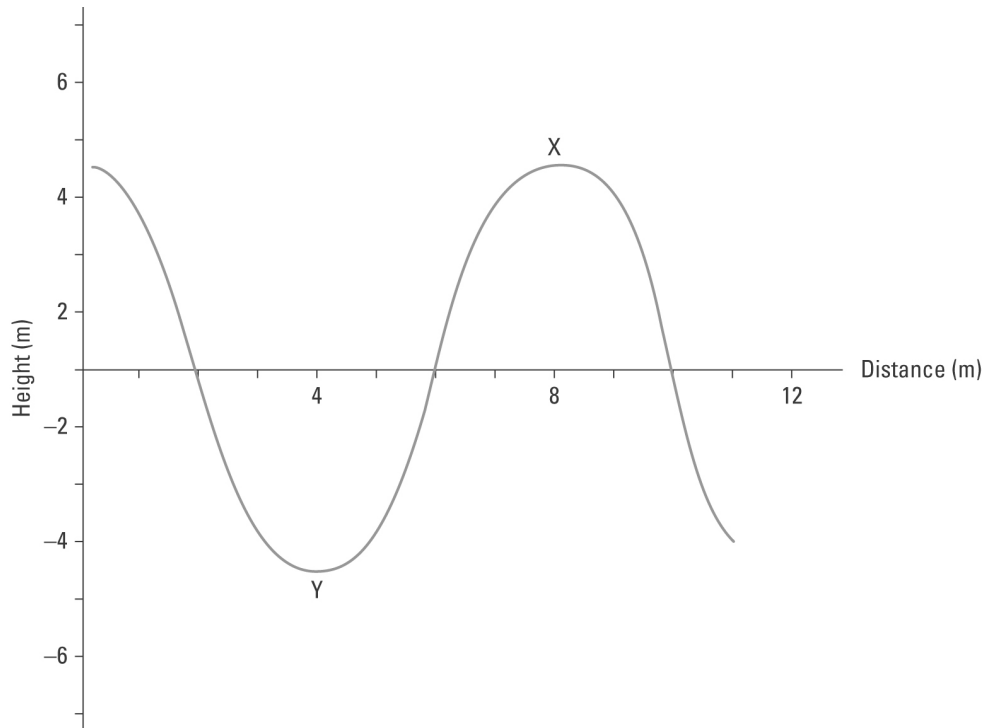


Waves

Student: Class:

1. Examine the following diagram of a wave.



- (a) Identify the type of wave illustrated.
- (b) Two parts of the wave are labelled X and Y. Name these parts.
X: Y:
.....
- (c) Use the scale to determine the wavelength and amplitude of the wave.
Wavelength:
Amplitude:
- (d) If the velocity of the wave is 16 m/s, calculate the frequency of the wave.

.....

.....

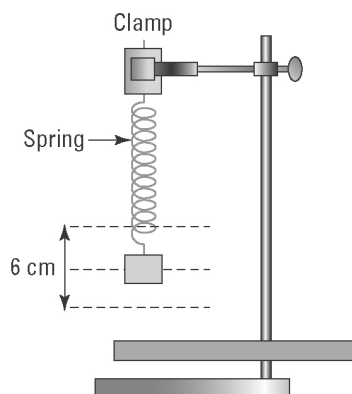
.....

.....

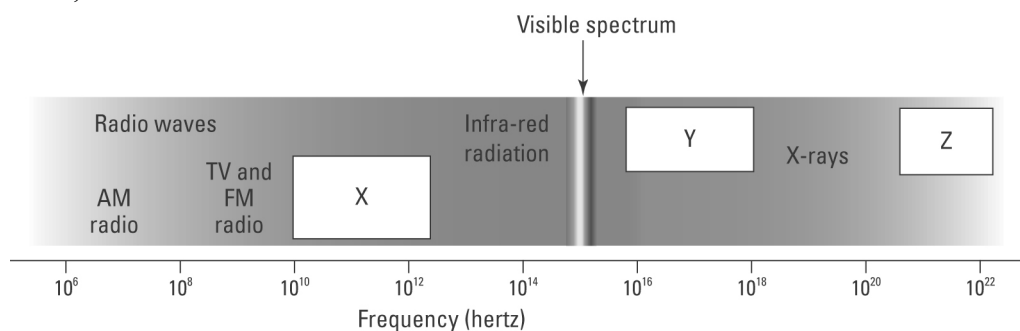
.....

.....

2. Consider the following diagram in which a mass is suspended from the end of a spring that is clamped to a retort stand.



- (a) The mass is pulled down and let go. It begins to oscillate up and down. Identify the type of wave motion in the spring.
-
- (b) Determine the amplitude of the vibration.
-
- (c) 20 oscillations are made in 5 seconds. Determine the frequency of the oscillation.
-
-
3. Electromagnetic waves are transverse waves which have very high frequencies. Light is an electromagnetic wave. Describe the nature of an electromagnetic wave.
-
-
-
-
4. The following diagram shows the regions of the electromagnetic spectrum. Identify the missing regions X, Y and Z.



X:

Y:

Z: